Supplementary Table 3 | Input and output tensor dimensions for each module.

Module	Input Size	Description	Output Size	Description
3D U-Net	2 × 256 × 256 × 32	Two consecutive 3D image frames	4 × 256 × 256 × 32	Confidence map (background/embryo /polar body/size)
			2 × 256 × 256 × 32	Division probability map (2-classes)
			64 × 256 × 256 × 32	Embedding map
CEN	6 × 256 × 256 × 32	Two frames + confidence map	5 × 256 × 256 × 32	Confidence map (center-enhanced, 4 levels + background)
Intra-frame MLP	64 × 1	Current cell embedding	6 × 1	5 candidates + 'None' scores
	64 × 5	Embeddings of 3candidates		
	3 × 5	Spatial distance to candidates		
	6 × 5	Difference and ratio of size and spatial distance		
	6 × 5	Difference and ratio of division		
Inter-frame MLP	64 × 1	Current cell embedding	6 × 1	5 candidates + 'None' scores
	64 × 5	Embeddings of 5 candidates		
	3 × 5	Spatial distance to candidates		
	6 × 5	Difference and ratio of size and spatial distance	2 × 1	Division probability
	6 × 5	Difference and ratio of division probability		